

NAME

CLASS

## ST ANDREW'S JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATIONS  
(WORKED SOLUTIONS)

Chemistry

8873/01

Higher 1

16 Sept 2021

Paper 1

1400 – 1500

1 hour

Additional materials: Data Booklet and Optical Answer Sheet

## READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and index number on the OAS provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.Choose the **one** you consider correct and record your choice in **soft pencil** on the separate OAS.**Read the instructions on the OAS very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of 21 pages.

- 1 The table shows the number of protons, neutrons and electrons present in some atoms and simple ions.

Which row describes two particles which are different isotopes of the same element with same charges?

|          | Particle 1 |          |           | Particle 2 |          |           |
|----------|------------|----------|-----------|------------|----------|-----------|
|          | protons    | neutrons | electrons | protons    | neutrons | electrons |
| <b>A</b> | 15         | 16       | 16        | 15         | 15       | 16        |
| <b>B</b> | 15         | 17       | 15        | 16         | 15       | 16        |
| <b>C</b> | 16         | 16       | 18        | 16         | 16       | 15        |
| <b>D</b> | 16         | 17       | 15        | 15         | 17       | 13        |

Ans: **A**

|          | Particle 1 |          |           |    | Particle 2 |          |           |    |
|----------|------------|----------|-----------|----|------------|----------|-----------|----|
|          | protons    | neutrons | electrons |    | protons    | neutrons | electrons |    |
| <b>A</b> | 15         | 16       | 16        | -1 | 15         | 15       | 16        | -1 |
| <b>B</b> | 15         | 17       | 15        | 0  | 16         | 15       | 16        | 0  |
| <b>C</b> | 16         | 16       | 18        | -2 | 16         | 16       | 15        | +1 |
| <b>D</b> | 16         | 17       | 15        | +1 | 15         | 17       | 13        | +2 |

Isotopes have the same number of protons but different number of neutrons. Only A has the same charges.

- 2 When 20 cm<sup>3</sup> of a hydrocarbon was burnt in 150 cm<sup>3</sup> of oxygen (an excess), the final gaseous mixture had a total volume of 110 cm<sup>3</sup>. On passing through aqueous potassium hydroxide, the volume of gaseous mixture decreased by 60 cm<sup>3</sup>. All gaseous volumes were measured under room temperature and pressure.

What is the formula of the hydrocarbon?

- A** C<sub>3</sub>H<sub>4</sub>  
**B** C<sub>3</sub>H<sub>8</sub>  
**C** C<sub>6</sub>H<sub>12</sub>  
**D** C<sub>6</sub>H<sub>14</sub>

Ans: **B**

Volume of C<sub>x</sub>H<sub>y</sub> = 20 cm<sup>3</sup>

Volume of O<sub>2</sub> unreacted = 110 – 60 = 50 cm<sup>3</sup>

Volume of O<sub>2</sub> reacted = 150 – 50 = 100 cm<sup>3</sup>

Volume of CO<sub>2</sub> = 60 cm<sup>3</sup>

C<sub>x</sub>H<sub>y</sub>: O<sub>2</sub> : CO<sub>2</sub> = 20 : 100 : 60 = 1 : 5 : 3

Hence,  $x = 3$

$(x+y/4 = 5 \rightarrow y = 8)$

- 3** When 0.2 mol of potassium chlorate,  $\text{KClO}_3$  is heated at its melting point, it forms potassium chlorate  $\text{KClO}_4$  and potassium chloride.

Which of the following statements is correct about this reaction?

- A** 0.2 mol of  $\text{KClO}_4$  is formed.
- B** It is a disproportionation reaction.
- C** Water is formed as a by-product of this reaction.
- D** The oxidation state of  $\text{Cl}$  in  $\text{KClO}_3$  is +7 while the oxidation of  $\text{Cl}$  in  $\text{KClO}_4$  is +5.

Ans: **B**



Option A:

No. of moles of  $\text{ClO}_3^- = 0.2 \text{ mol}$

No. of moles of  $\text{ClO}_4^- = 0.2/4 \times 3 = 0.15 \text{ mol}$

Option B: This is a disproportionation reaction because the oxidation state of  $\text{Cl}$  in  $\text{KClO}_3$  increases from +5 to +7 in  $\text{KClO}_4$  and decrease to  $-1$  in  $\text{KCl}$ .

Option C: Water is not formed as a by-product.

Option D:

The oxidation state of  $\text{Cl}$  in  $\text{KClO}_3$  is +5 while the oxidation of  $\text{Cl}$  in  $\text{KClO}_4$  is +7.

- 4** Which one of the following ions will be deflected to the greatest extent in an electric field?

- A**  $^{23}\text{Na}^+$
- B**  $^{27}\text{Al}^{3+}$
- C**  $^{32}\text{S}^{2-}$
- D**  $^{14}\text{N}^{3-}$

Ans: **D**

N has the greatest charge and smallest mass, hence experience the greatest deflection.

- 5 The first eight ionisation energies of an element **Q** (in  $\text{kJ mol}^{-1}$ ) are as follows:

966, 1950, 2730, 4850, 6020, 12300, 15400, 18900

Which of the following statements about element **Q** is correct?

- A It has an even number of protons in its atoms.
- B It can form a chloride with the formula  $\text{QCl}_5$ .
- C It can form a chloride that is trigonal planar in shape.
- D It can form a basic oxide.

Ans: **B**

The increase between the fifth and sixth IE is the largest. Hence, there is 5 valence electrons. Hence, the formula is  $\text{QCl}_5$ .

An acidic oxide is formed.

- 6 Why is the first ionisation of sulfur lower than that of phosphorus?

- A Sulfur is more electronegative than phosphorus.
- B Sulfur forms anions more readily.
- C The electron to be lost in sulfur is paired with another electron.
- D The sulfur atom is larger than the phosphorus atom.

Ans: **C**

The electronic configuration of S is  $1s^2 2s^2 2p^6 3s^2 3p^4$ . There is inter-electronic repulsion between the 3 p electrons in the doubly filled 3 p orbital. Hence, less energy is required to remove the paired p orbitals.

- 7 Which of the following descriptions of polarity and bond angle of the molecule is correct?

|          | Molecule        | Polarity  | Bond angle    |
|----------|-----------------|-----------|---------------|
| <b>1</b> | $\text{BeCl}_2$ | polar     | $104.5^\circ$ |
| <b>2</b> | $\text{SiCl}_4$ | non-polar | $109.5^\circ$ |
| <b>3</b> | $\text{XeCl}_4$ | non-polar | $90^\circ$    |

- A** 1 and 2 only  
**B** 2 and 3 only  
**C** 2 only  
**D** 3 only

Ans: **B**

|          | Molecule        | Polarity  | Bonding                        |
|----------|-----------------|-----------|--------------------------------|
| <b>1</b> | $\text{BeCl}_2$ | non-polar | $180^\circ$<br>(linear)        |
| <b>2</b> | $\text{SiCl}_4$ | non-polar | $109.5^\circ$<br>(tetrahedral) |
| <b>3</b> | $\text{XeCl}_4$ | non-polar | $90^\circ$<br>(square planar)  |

- 8 Which of the following molecules required the highest pressure to be liquefied?

- A**  $\text{CH}_4$   
**B**  $\text{NH}_3$   
**C**  $\text{HF}$   
**D**  $\text{H}_2\text{O}$

Ans: **A**

$\text{CH}_4$  is a tetrahedral non-polar simple covalent molecule which has id-id interaction, the weakest intermolecular attraction among the options. Hence, it would require the highest pressure to liquefy.

- 9 When 25 cm<sup>3</sup> of 2.0 mol dm<sup>-3</sup> sodium hydroxide is completely neutralised by 25 cm<sup>3</sup> of 3.0 mol dm<sup>-3</sup> hydrochloric acid, the temperature of the mixture rose by 6.8 °C. Assuming negligible heat loss, what is the enthalpy of neutralisation in kJ mol<sup>-1</sup> for this reaction?
- A + 14.2  
 B + 18.9  
 C - 14.2  
 D - 28.4

Ans: D

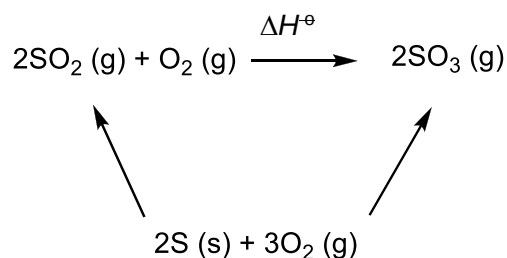
No. of moles of NaOH = No. of moles of H<sub>2</sub>O = 0.05 mol

(NaOH is the limiting reagent)

$$Q = mc\Delta T = 50 \times 4.18 \times 6.8 = 1421.2 \text{ J}$$

$$\Delta H = -1421.2 / 0.05 = -28424 \text{ J mol}^{-1} = -28.4 \text{ kJ mol}^{-1}$$

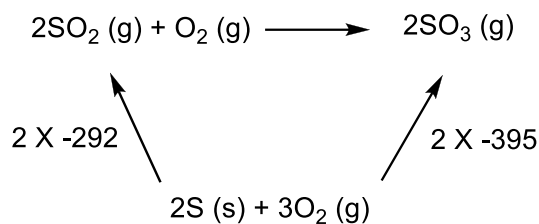
- 10 Given the following enthalpy changes:



Using the energy cycle above, what is the enthalpy change of reaction,  $\Delta H^\ominus$  in kJ mol<sup>-1</sup>?

- A +189  
 B -687  
 C -206  
 D -1374

Ans: C



By Hess' law,  $\Delta H = 292 \times 2 + (2 \times -395) = -206 \text{ kJ mol}^{-1}$

- 11** At 25°C, sucrose decomposes in acidic solution into glucose and fructose according to the rate equation below:

$$\text{Rate} = k [\text{sucrose}]$$

A sample of sucrose was allowed to decompose in acidic solution. At the end of 24 hours, it was found that only 12.5% of the original sample was left in the reaction flask. What is the half-life of the decomposition of sucrose in acidic solution?

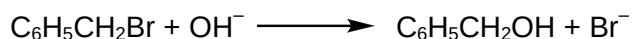
- A** 8 hours
- B** 12 hours
- C** 24 hours
- D** 72 hours

Ans: **A**

The rate order wrt [sucrose] is 1, hence, half life is constant.

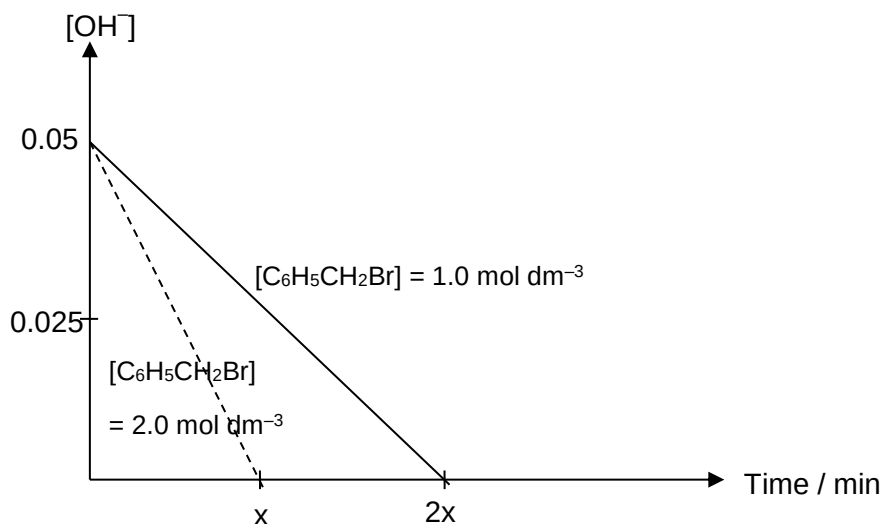
12.5 % means that 1/8 of sucrose remains. That would mean that 3 half lives have passed. Hence, half life of sucrose =  $24/3 = 8$  hours.

12



The reaction kinetics of the above reaction is determined by monitoring the change in the concentration of  $\text{OH}^-$  with time.

The results are shown below:



What is the rate equation for this reaction?

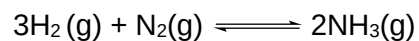
- A**    rate =  $k[\text{C}_6\text{H}_5\text{CH}_2\text{Br}]$
- B**    rate =  $k[\text{OH}^-]$
- C**    rate =  $k[\text{C}_6\text{H}_5\text{CH}_2\text{Br}]^2$
- D**    rate =  $k[\text{C}_6\text{H}_5\text{CH}_2\text{Br}][\text{OH}^-]$

Ans: **A**

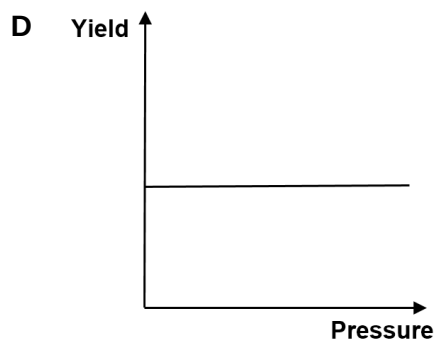
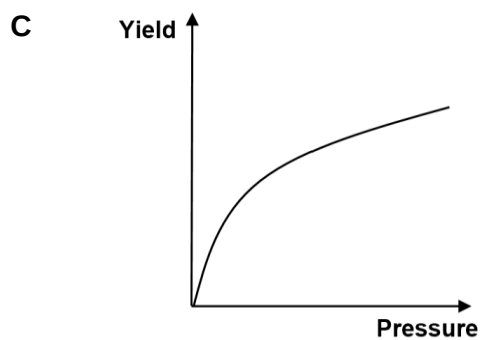
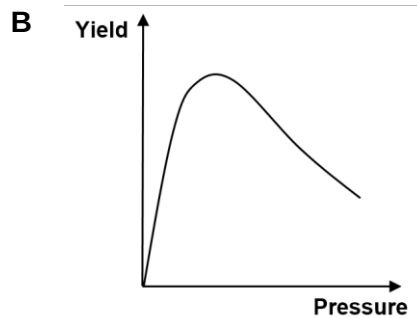
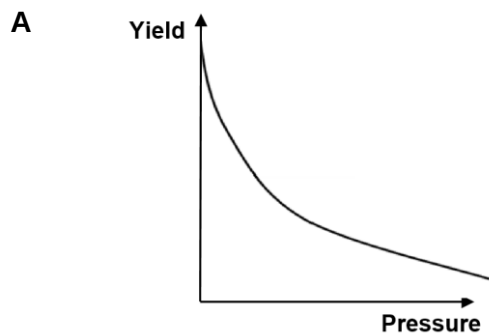
The downward straight line graph indicates that the rate order wrt  $[\text{OH}^-]$  is zero. The gradient is twice (change of y is the same while change of x is halved) when the  $[\text{C}_6\text{H}_5\text{CH}_2\text{Br}]$  is twice. Hence, the rate order wrt  $[\text{C}_6\text{H}_5\text{CH}_2\text{Br}]$  is one.



- 13** Ammonia can be synthesised directly using a mixture of hydrogen gas and nitrogen gas in a reversible reaction as shown:



Which one of the following describes the change in yield of ammonia at equilibrium as pressure increases?



Ans: **C**

When pressure increases, POE shifts to RHS to favour less number of moles of  $\text{NH}_3$  to remove the pressure. Hence, the yield increases.

**14** Which of the following mixtures are a *Bronsted-Lowry* acid and conjugate base pair?

- 1**  $\text{H}_2\text{O}$  and  $\text{OH}^-$
- 2**  $\text{H}_2\text{PO}_4^-$  and  $\text{HPO}_4^{2-}$
- 3**  $\text{NaH}$  and  $\text{Na}$

- A** 1, 2 and 3 only
- B** 1 and 2 only
- C** 2 only
- D** 3 only

Ans: **B**

A Bronsted Lowry acid, HA is a proton donor. It donates its proton to be a conjugate base  $\text{A}^-$ .

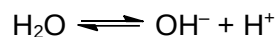
Hence,

1: correct

2: correct

3: not an acidic dissociation ( $\text{NaH}$  is not a weak acid)

**15** The value of the ionic product of water,  $K_w$ , varies with temperature.



| temperature / °C | $K_w / \text{mol}^2 \text{dm}^{-6}$ |
|------------------|-------------------------------------|
| 0                | $0.1 \times 10^{-14}$               |
| 10               | $0.3 \times 10^{-14}$               |
| 25               | $1.0 \times 10^{-14}$               |

Which row is correct?

- |          |                            |                                     |
|----------|----------------------------|-------------------------------------|
|          | The ionisation of water is | At 25 °C, water with a pH of 7.0 is |
| <b>A</b> | endothermic                | alkaline                            |
| <b>B</b> | endothermic                | neutral                             |
| <b>C</b> | exothermic                 | alkaline                            |
| <b>D</b> | exothermic                 | neutral                             |

Ans: **B**

When temperature increase,  $K_w$  increases. This shows that the forward reaction is endothermic. Hence, the POE shifts to the RHS, more  $\text{H}^+$  produced. The amount of  $\text{H}^+$  is the same as  $\text{OH}^-$ , hence it is still a neutral liquid.

- 16** Solutions **E**, **F**, **G** and **H** contain either a strong or weak acid or base, but not necessarily in the same order. The concentration and pH for each solution are shown below.

| Solution | Concentration /mol dm <sup>-3</sup> | pH at 25°C |
|----------|-------------------------------------|------------|
| <b>E</b> | 1.0                                 | 4.0        |
| <b>F</b> | 1.0                                 | 9.4        |
| <b>G</b> | 0.01                                | 12.0       |
| <b>H</b> | 0.01                                | 2.0        |

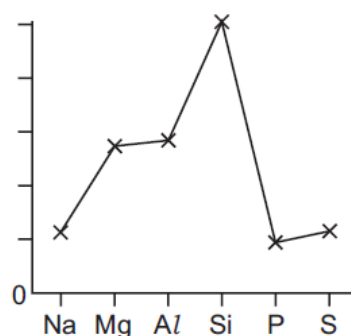
Which one of the following statements is a weak monobasic acid?

- A**      **E**  
**B**      **F**  
**C**      **G**  
**D**      **H**

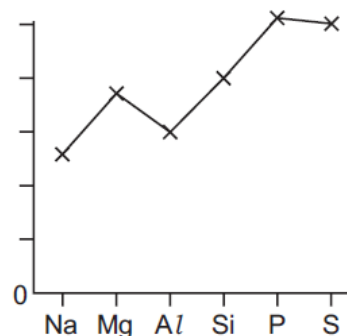
Ans: **A**

| Solution | Concentration /mol dm <sup>-3</sup> | pH   | Type        |
|----------|-------------------------------------|------|-------------|
| <b>E</b> | 1.0                                 | 4.0  | Weak acid   |
| <b>F</b> | 1.0                                 | 9.4  | Weak base   |
| <b>G</b> | 0.01                                | 12.0 | Strong Base |
| <b>H</b> | 0.01                                | 2.0  | Strong Acid |

- 17 The trends in three physical properties of the elements Na, Mg, Al, Si, P and S are shown in the following graphs.



Graph 1



Graph 2

Which of the following illustrate the correct physical property for the corresponding graphs above?

- |   | Graph 1                           | Graph 2                           |
|---|-----------------------------------|-----------------------------------|
| A | melting point                     | electrical conductivity           |
| B | 1 <sup>st</sup> ionisation energy | electrical conductivity           |
| C | melting point                     | 1 <sup>st</sup> ionisation energy |
| D | 1 <sup>st</sup> ionisation energy | melting point                     |

Ans: C

Graph 1: The melting point increases from Na to Al as the metallic bonding is stronger. Si has the highest melting point as it is a giant covalent lattice which requires the largest amount of energy to overcome the strong network of Si-Si bond. S<sub>8</sub> and P<sub>4</sub> are simple covalent molecule. As the number of electrons for S<sub>8</sub> increases, the intermolecular forces increase, hence, the melting point increases.

Graph 2: Across the period, nuclear charge increases while the increase in shielding effect is insignificant as electrons are added to the same shell. The effective nuclear charge increases, hence, the 1<sup>st</sup> ionisation energy increases. However, the first ionisation energy for Al is lower than expected because less energy is needed to remove the 3p electron which is further away from nucleus and experience additional shielding from 3s electrons.

- 18** Three samples of white powder, **X**, **Y** and **Z** were found. The possible identities of each sample are  $\text{Al}_2\text{O}_3$ ,  $\text{P}_4\text{O}_{10}$  and  $\text{SiO}_2$ . Aqueous hydrochloric acid and sodium hydroxide were added to test their identities.

|          | HCl (aq)    | NaOH (aq)   |
|----------|-------------|-------------|
| <b>X</b> | No reaction | Reaction    |
| <b>Y</b> | Reaction    | Reaction    |
| <b>Z</b> | No reaction | No reaction |

Which of the following shows the correct identities of **X**, **Y** and **Z**?

|          | <b>X</b>                  | <b>Y</b>                  | <b>Z</b>                  |
|----------|---------------------------|---------------------------|---------------------------|
| <b>A</b> | $\text{Al}_2\text{O}_3$   | $\text{P}_4\text{O}_{10}$ | $\text{SiO}_2$            |
| <b>B</b> | $\text{P}_4\text{O}_{10}$ | $\text{SiO}_2$            | $\text{Al}_2\text{O}_3$   |
| <b>C</b> | $\text{P}_4\text{O}_{10}$ | $\text{Al}_2\text{O}_3$   | $\text{SiO}_2$            |
| <b>D</b> | $\text{SiO}_2$            | $\text{Al}_2\text{O}_3$   | $\text{P}_4\text{O}_{10}$ |

Ans: **C**

|                           | HCl (aq)                                              | NaOH (aq)                                                 |
|---------------------------|-------------------------------------------------------|-----------------------------------------------------------|
| $\text{P}_4\text{O}_{10}$ | No reaction                                           | $\text{P}_4\text{O}_{10}$ is acidic and reacts with base. |
| $\text{Al}_2\text{O}_3$   | Amphoteric in nature, hence reacts with acid and base |                                                           |
| $\text{SiO}_2$            | No reaction                                           | No reaction, except concentrated NaOH with heat           |

- 19** Which of the following statements about the trends in the properties of halogens are correct on descending the group?

- The electronegativity decreases.
- The volatility increases.
- Their reactivity as oxidising agents decreases.

- A** 1, 2 and 3 only  
**B** 2 and 3 only  
**C** 1 and 3 only  
**D** 2 only

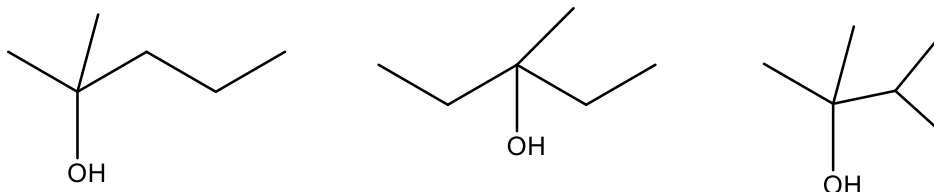
Ans: **C**

The halogen becomes less volatile because of stronger id-id interaction formed as the electron cloud increases.

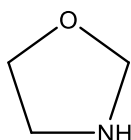
20 How many tertiary alcohols have the molecular formula  $C_6H_{14}O$ ?

- A 1
- B 2
- C 3
- D 4

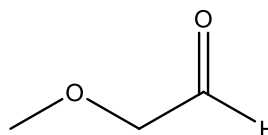
Ans: C



21 Which one of the following statements best accounts for the difference in the boiling points of oxazolidine and compound **W**?



Oxazolidine



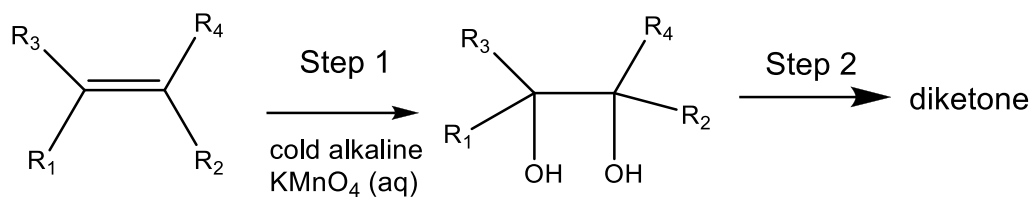
Compound **W**

- A Oxygen is more electronegative than nitrogen.
- B Oxazolidine has a smaller surface area than **W** for intermolecular interactions.
- C Oxazolidine has 3 lone pairs of electrons while **W** has 4 lone pairs of electrons.
- D Hydrogen bonding present between oxazolidine molecules is stronger than the permanent dipole- permanent dipole interactions present between **W** molecules.

Ans: D

As explained in Option D.

The following reaction scheme occurs in an alkene. It is relevant for Question **22** and **23**.



**J**

$\text{R}_1$ ,  $\text{R}_2$ ,  $\text{R}_3$  and  $\text{R}_4$  represent alkyl groups or H.

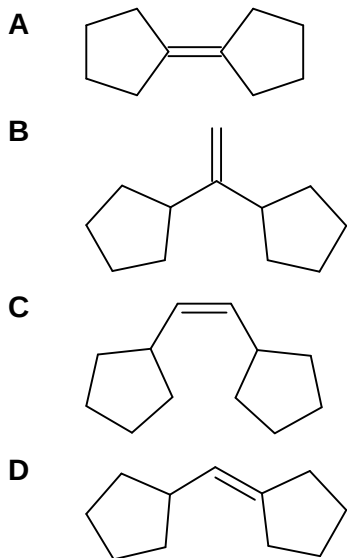
**22** What types of reaction occur in Step 1 and Step 2?

|          | Step 1       | Step 2    |
|----------|--------------|-----------|
| <b>A</b> | addition     | addition  |
| <b>B</b> | oxidation    | oxidation |
| <b>C</b> | substitution | addition  |
| <b>D</b> | oxidation    | reduction |

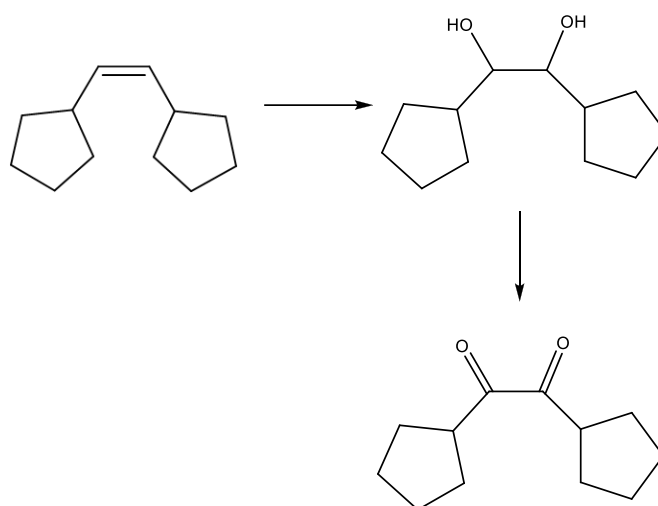
Ans: **B**

Cold alkaline  $\text{KMnO}_4$  (aq) is an oxidising agent. Hence oxidation takes place. The conversion of alcohol to ketone is an oxidation process.

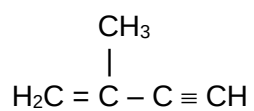
**23** What is a possible identity of the starting material, **J** to give a diketone?



Ans: **C**



- 24 Which of the following options shows the correct number and type of bonds in Molecule K?



Molecule K

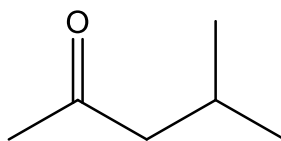
|          | Number of $\sigma$ - bonds | Number of $\pi$ - bonds |
|----------|----------------------------|-------------------------|
| <b>A</b> | 4                          | 3                       |
| <b>B</b> | 8                          | 5                       |
| <b>C</b> | 10                         | 2                       |
| <b>D</b> | 10                         | 3                       |

Ans: **D**

10 sigma + 3 pi (1 from C double bond and 1 from C triple bond)



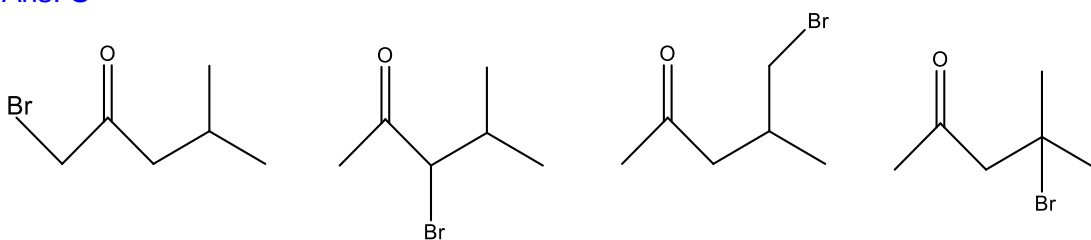
- 25 How many different mono-brominated compounds can be formed when Compound **H** reacts with gaseous bromine under the presence of UV light?



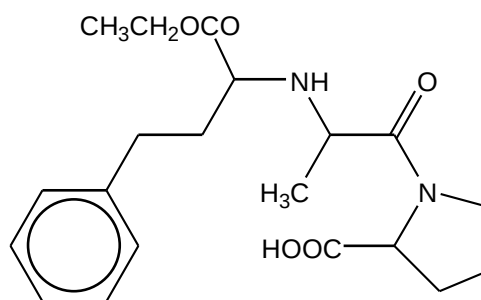
Compound **H**

- A 2  
B 3  
C 4  
D 5

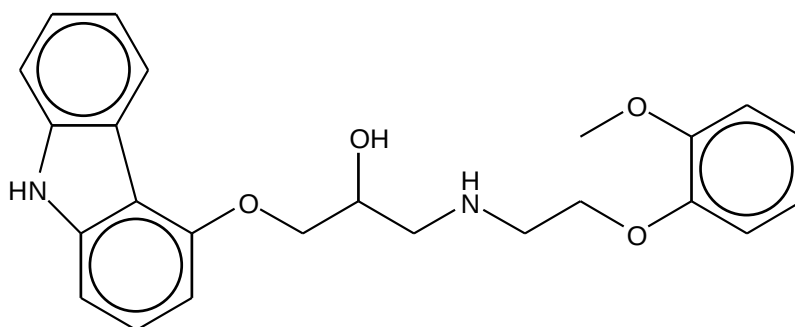
Ans: **C**



- 26** Enalapril and Carvedilol are two compounds that are used in the treatment of heart failure and disease.



**Enalapril**



**Carvedilol**

Which of the following can be used to distinguish between the two compounds?

- 1  $\text{Na}_2\text{CO}_3$  (aq)
- 2  $\text{H}_2/\text{Pt}$
- 3 Ethanolic KOH

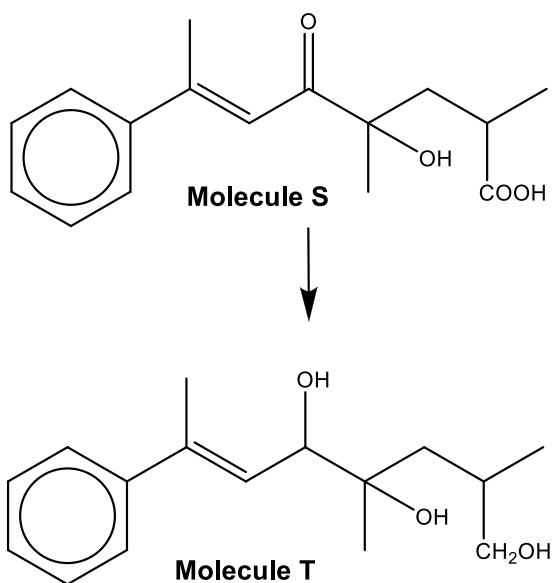
- A** 1 only  
**B** 1 and 2 only  
**C** 2 and 3 only  
**D** 1 and 3 only

Ans: **A**

Enalapril has  $-\text{COOH}$  which can react with  $\text{Na}_2\text{CO}_3$  (acid-carbonate reaction) while no  $\text{CO}_2$  is produced in Carvedilol.

No reaction occurs with both compounds with  $\text{H}_2/\text{Pt}$  and ethanolic KOH.

27 Which of the following reducing agents would convert Molecule **S** to Molecule **T**?



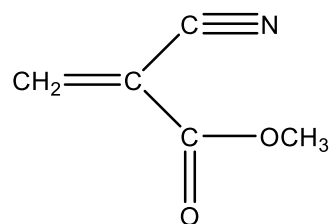
- 1  $\text{LiAlH}_4$ / dry ether
- 2  $\text{NaBH}_4$
- 3  $\text{H}_2$ / Ni

- A 1, 2 and 3 only
- B 1 and 2 only
- C 1 only
- D 2 only

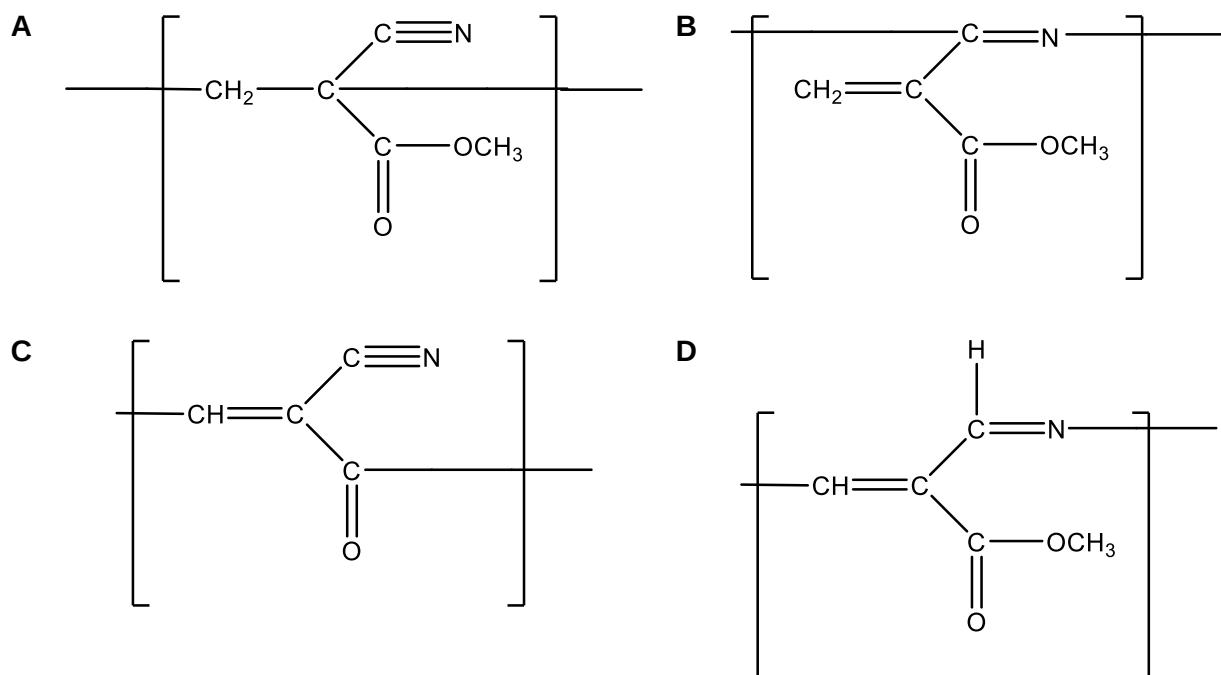
Ans: **C**

If  $\text{H}_2/\text{Ni}$  is used, the double bond will be reduced too.

28 "Superglue" contains the following compound



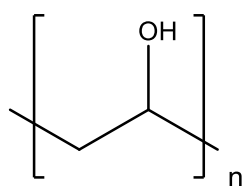
It is rapidly polymerised by traces of base on the surface of the objects to be stuck together. Which of the following represents the repeat unit of the polymerised form?



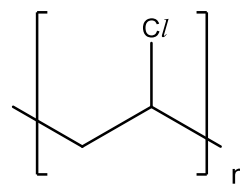
Ans: **A**

Addition polymerization occurs at the  $-\text{C}=\text{C}-$  groups.

- 29 Which of the following about polyvinylalcohol (PVA) and polyvinylchloride (PVC) is correct?



**PVA**



**PVC**

- A** PVC is a poorer water resistant polymer than PVA.
- B** PVC can react with NaOH via a substitution reaction
- C** Both PVC and PVA has hydrogen bonding present between its polymeric chains.
- D** PVA is used in eye drops because of the strong permanent dipole – permanent dipole interaction with water molecules.

Ans: **B**

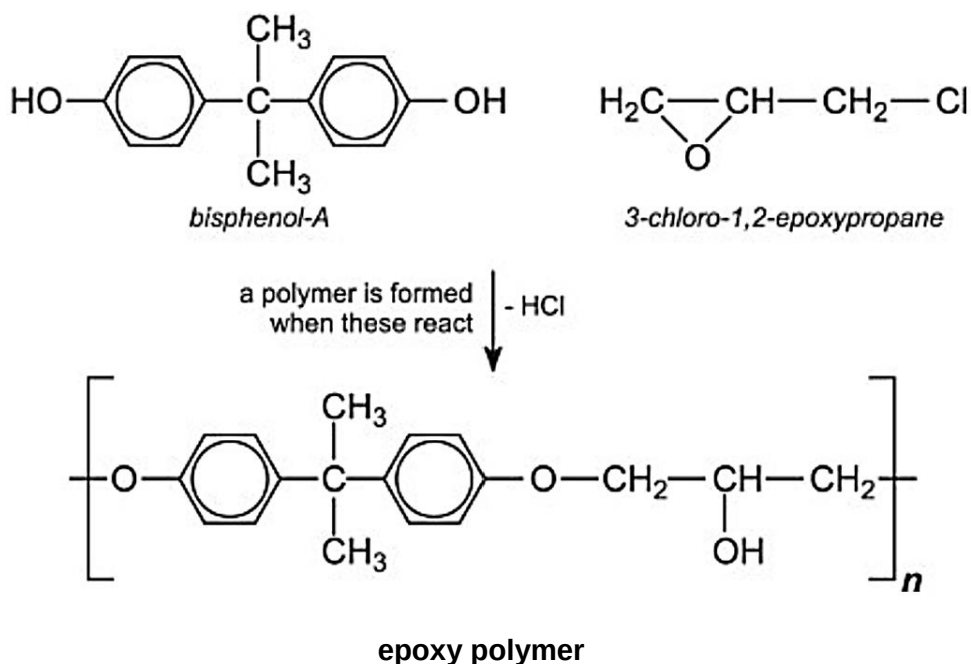
Option A: PVC is a better water resistant polymer as it forms pd-pd between PVC chains and water molecules which does not evolve enough energy to cover pd-pd between PVC chains and hydrogen bonding between polymer chains.

Option B: Correct. ( $\text{RCI} + \text{NaOH} \rightarrow \text{ROH} + \text{HCl}$ ) – substitution

Option C: PVC has permanent dipole-permanent dipole interaction between its polymeric chains while PVA has hydrogen bonding between its polymeric chains.

Option D: PVA is used in eye drops because of the strong hydrogen bonding interaction with water molecules.

30 Epoxy polymers can be made as shown:



Source: <https://www.essentialchemicalindustry.org/materials-and-applications/paints.html>

Which of the following statements is most likely to be **false** about epoxy polymer?

- A It is a condensation polymer.
- B It does not form cross-links with its own polymeric chains.
- C It is water-soluble.
- D It does not degrade via hydrolysis.

Ans: **C**

Option A: It is a condensation polymer because HCl is formed when bisphenol-A and 3-chloro-1,2-epoxypropane reacts.

Option B: It does not have  $\text{C}=\text{C}$  or both unreacted acid/amine/alcohol to form cross-links.

Option C: Although it is capable of forming hydrogen bonding with water molecule, however, the large benzene ring are hydrophobic, the hydrophobic interaction cannot be easily overcome by hydrogen bonding with water.

Option D: There is no presence of ester or amide that can undergo hydrolysis.